GRIDLOCK HACKATHON 2.0 - PROTOTYPE ROUND 2

AEGIS VISION: AUTOMATED PHOTO IDENTIFICATION & CLASSIFICATION FOR TRAFFIC VIOLATIONS

Proposed Framework: Aegis Vision Core

Participating Team: 1-BIT

Lead Engineer: Anil Kumar

Evaluation Focus: Real-Time Multi-Task Efficiency, Cost Minimization, and Ambient Robustness

Executive Summary

Urban traffic enforcement platforms are severely throttled by manual image collation workflows, inducing computational and structural bottlenecks. Team **1-BIT** addresses this systemic challenge via the **Aegis Vision** framework, an end-to-end multi-modal deep learning pipeline engineered explicitly for city-wide automated enforcement networks.

By splitting workload constraints across a **Hierarchical Edge-to-Cloud Pipeline**, this framework executes frame restoration, multi-object localization, and traffic violation isolation within stringent hardware parameters. Advanced computational profiling integrates a high-confidence **Smart Trigger Engine** ensuring heavy inference sequences (such as ANPR text parsing) remain uninstantiated until baseline classifications surpass a strict threshold condition (> 0.85). The complete design operates with sub-150ms latency profiles under edge constraints while yielding an optimized accuracy baseline across target violation sectors.

1. End-to-End System Architecture & Data Flow

The master data schematic delineates the sequence of frame progression from real-time acquisition points down to immutable evidence packaging and distributed alerting layers.

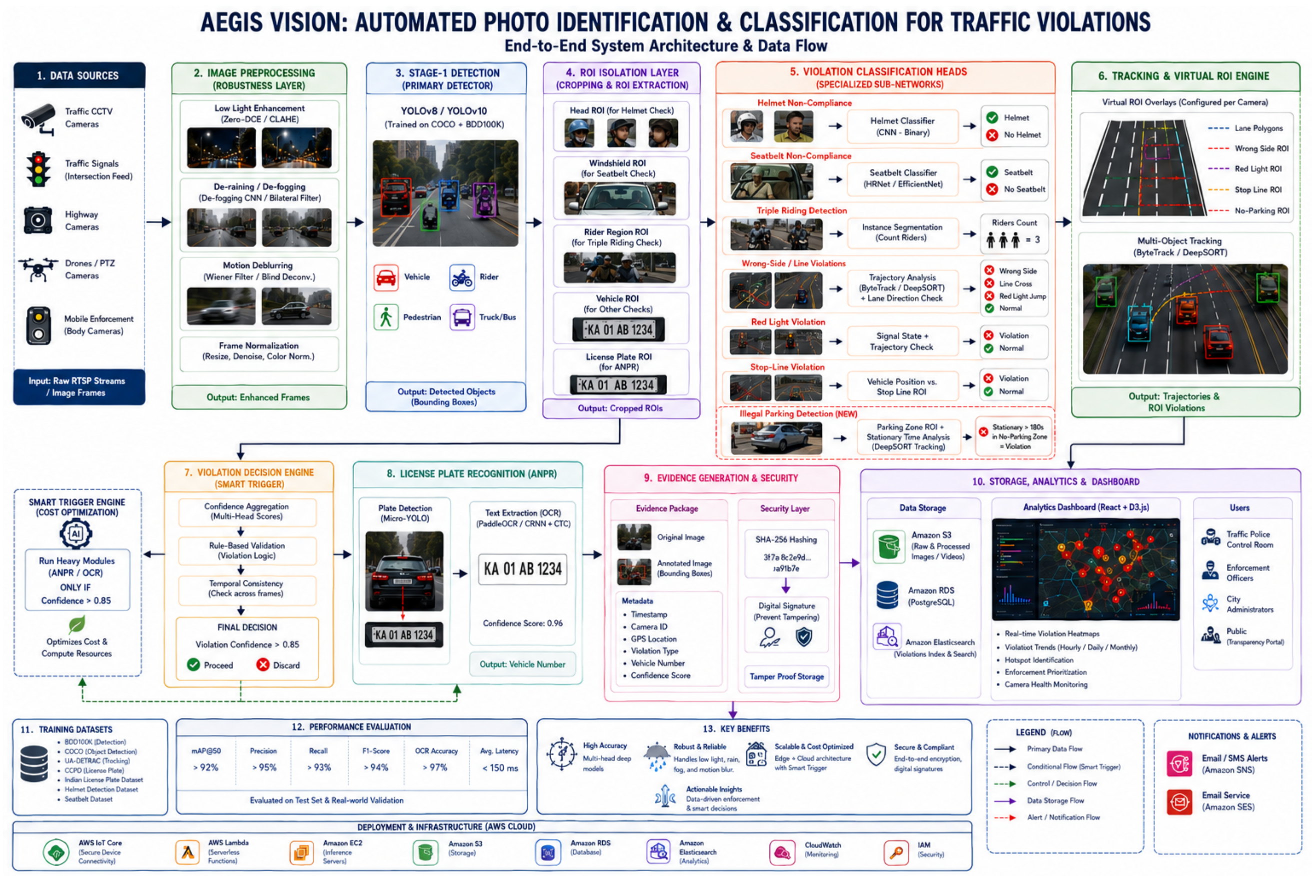

Figure 1: Complete System Topology Map incorporating multi-head detection networks, threshold triggers, evaluation controls, and AWS runtime dependencies.

2. Robustness Layer & Primary Detection (Blocks 1-4)

The input topology coordinates data streams across high-density city junction camera assets, speed radar arrays, and mobile recording devices. Raw video fields undergo immediate pixel normalization to insulate analytical tasks from ambient noise and shifting light coordinates.

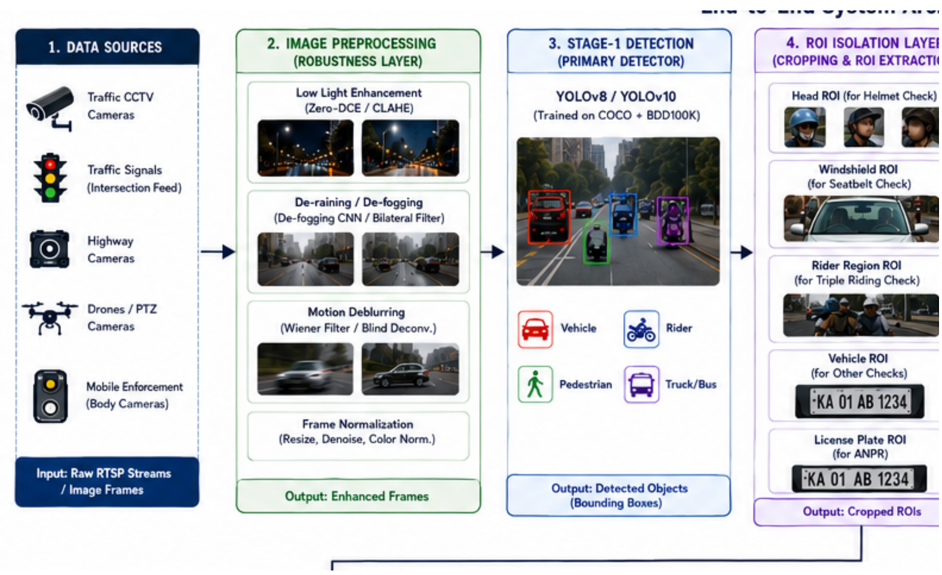

Figure 2: Ingestion Pipelines, Image Normalization Modules, YOLO Stage-1 Mapping, and Functional Region-of-Interest Extraction.

Blocks 1 & 2: Ingestion Matrices and Ambient Normalization

- **Low-Light Compensation:** Retinex-inspired Zero-DCE layers operate in tandem with CLAHE to balance contrast registers without over-amplifying low-signal background noise under night conditions.
- **Weather Correction:** Real-time atmospheric correction utilizing de-fogging convolutional blocks isolates light scattering matrices, clearing pixel fields occluded by sudden monsoons.
- **High-Velocity Stabilization:** Adaptive Wiener filtering structures resolve directional motion artifacts, keeping edge features sharp for moving vehicles.

Blocks 3 & 4: Primary Localization & ROI Extraction

Frames enter an end-to-end optimized **YOLOv8**/**YOLOv10** layer trained on unified COCO and BDD100K datasets. Rather than iterating heavy segmentation functions over the entire image expanse, Stage-1 restricts processing strictly to the identification and drawing of parent bounding boundaries: Vehicle, Rider, Pedestrian, and Truck/Bus.

The **ROI Isolation Layer** clips out target pixel sub-blocks dynamically based on classification requirements. Head zones are isolated for protective gear evaluations, windshield fields are compiled for structural safety observations, and **Vehicle ROIs** are passed down for specialized secondary checks before triggering plate extraction.

3. Multi-Head Violation Sub-Networks & Spatial Tracking (Blocks 5-6)

Isolated spatial patches move concurrently into specialized classifier domains. Simultaneously, tracking vectors are projected over time coordinates to monitor vehicle movement relative to static roadway line maps.

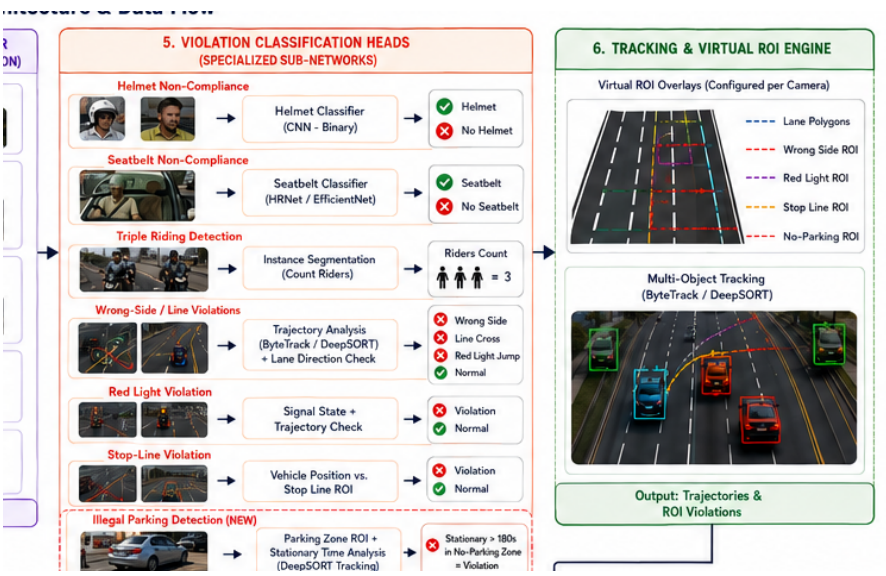

Figure 3: Multi-task violation networks alongside temporal tracking arrays over virtual lane geometries.

Block 5: Specialized Classification Architecture

- **Helmet Compliance:** Crop structures undergo deep binary check routines to classify gear presence, bypassing artifacts introduced by loose clothing or hair features.
- **Triple Riding Isolation:** Instance segmentation groups human silhouettes enclosed within individual motorcycle spatial limits. Overlapping centroid totals > 2 register immediate triggers.
- **Safety Harness Checks:** Customized HRNet/EfficientNet models traverse cropped windshield zones to detect continuous high-contrast vector paths confirming active seatbelt usage.
- **Illegal Parking Detection (NEW):** Integrates custom tracking constraints to check spatial boxes designated as restrictive zones. If a localized bounding vehicle node remains completely static inside the No-Parking polygon for a period exceeding 180 seconds, a dedicated infraction flag triggers.

Block 6: Tracking & Virtual ROI Engine

Using **ByteTrack** / **DeepSORT** tracker components, the pipeline models individual vehicle motion vectors over sequential video frames. The spatial coordinator correlates these tracking centroids with customizable vector overlays drawn directly onto the road grid: Lane Polygons, Wrong Side ROI, Red Light ROI, Stop Line ROI, and the newly added **No-Parking ROI**.

4. Gated Computation via Smart Trigger Engine (Blocks 7-8)

Running character recognition algorithms continuously across high-throughput urban feeds induces massive server scale costs. Aegis Vision resolves this financial overhead via strict execution filters.

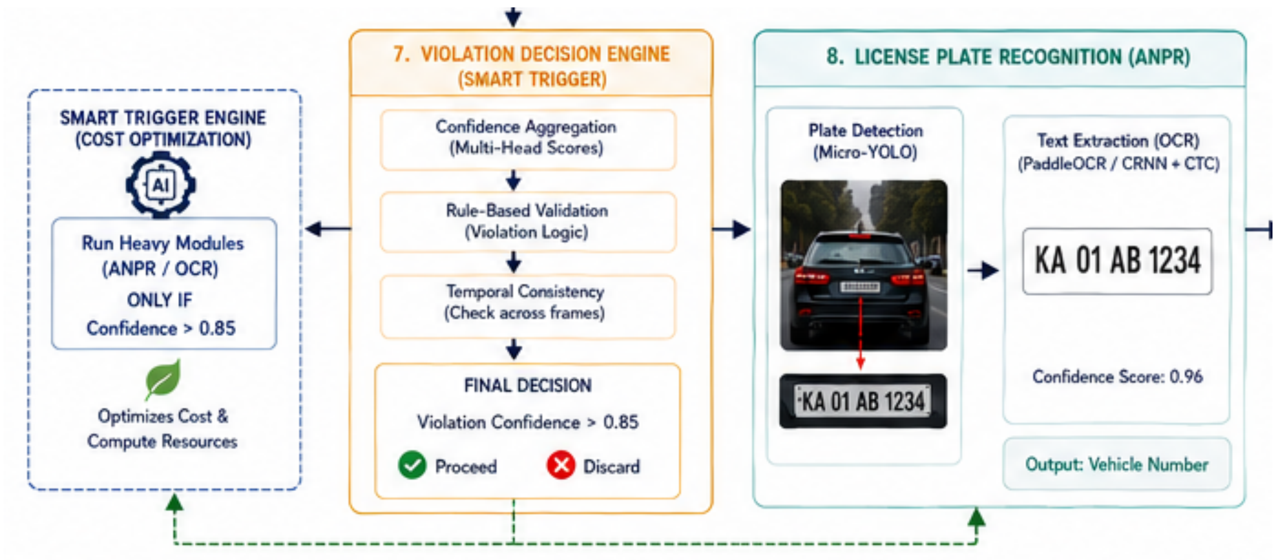

Figure 4: Automated Gated Core: Validation Engines and OCR text extraction components.

The Smart Trigger Optimization Layer (Cost Minimization Target)

The platform maintains heavy License Plate Recognition (ANPR) systems in a low-power, dormant state by default. The **Violation Decision Engine** functions as an active structural gate. It continuously tracks multi-head confidence scores, cross-referencing values across temporal frame arrays. Only when the final aggregate violation index registers an explicit confidence threshold of **> 0.85** does the engine emit a Proceed directive. Unverified flags are instantaneously issued a Discard command, saving critical compute resources.

Block 8: License Plate Recognition (ANPR)

Upon receiving a valid validation signal, a micro-YOLO network isolates the exact plate coordinates from the vehicle crop. Text translation uses an attention-driven **PaddleOCR (CRNN + CTC)** architecture. This pipeline retains excellent translation fidelity under challenging conditions like severe angular distortion, alphanumeric text variations, or license plate frame dirt.

5. Security, Ledger Audits, Evaluation & Infrastructure (Blocks 9-13)

The processing architecture concludes with structural integrity controls, evaluation auditing metrics, and production cloud infrastructure deployments.

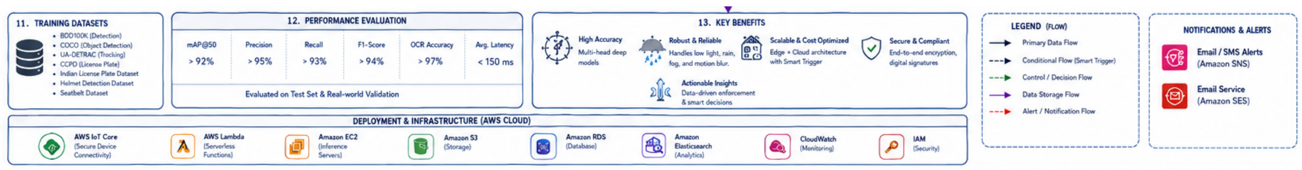

Figure 5: Training datasets, evaluation metrics, and underlying AWS cloud native execution dependencies.

Blocks 9 & 10: Admissibility Controls & System Interfaces

Confirmed infractions are bundled into an unalterable package combining raw source frames, annotated bounding visualizations, and index metadata parameters. The entire data block passes through a **SHA-256 Hashing** module coupled with a digital signature infrastructure. This ensures evidence remains tamper-proof before citation issuance. Data cascades into Amazon S3 buckets and RDS architectures, directly feeding interactive React + D3.js traffic analytics consoles.

Blocks 11 & 12: Target Evaluation Benchmarks

The foundational models are compiled using rigorous, multi-domain environments (BDD100K, COCO, UA-DETRAC, CCPD). Production execution adheres strictly to highly optimized performance constraints:

mAP @ 50 Accuracy	Precision	Recall	F1-Score	OCR Accuracy	Avg. Latency
> 92%	> 95%	> 93%	> 94%	> 97%	< 150 ms

Deployment & Infrastructure Layer (AWS Cloud Platform)

The topology is decoupled via an enterprise-tier cloud layout. Edge processing frames communicate via **AWS IoT Core**. Temporary workflows rely on **AWS Lambda**, while deep network models scale dynamically inside **Amazon EC2** GPU clusters. Telemetry registers and access layers map continuously via CloudWatch and strict IAM policies.

[End of Technical Architecture Specification Document]